Piter Z. Garcia Bautista

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Research Profile

Graduate researcher focused on machine learning for diagnostics and decision support under noisy, heterogeneous, and partially observed data. Background includes an M.S. in Computer Science, current M.S. work in Data Science, industry healthcare AI experience, and computational biology training. Current work emphasizes reproducibility, robust evaluation, fairness-aware analysis, and translational relevance to clinical settings.

Research Interests

Machine learning for precision medicine and diagnostics; robust learning under missing or uncertain context; fairness-aware and trustworthy AI for high-stakes decision support; reproducible computational evaluation systems.

Education

Rochester Institute of Technology — Expected Aug 2026
Master of Science, Data Science — GPA 3.9 — *Rochester, NY*

- Concentration: Bioinformatics & Artificial Intelligence.
- Relevant coursework: Bioinformatics Algorithms, Data-Driven Knowledge Discovery, Neural Networks, Applied Statistics, Software Engineering for Data Science.

University of Rochester, Warner School of Education — In progress
Graduate study, Teaching Computer Science K–12 (M.S. track) — GPA 4.0 — *Rochester, NY*

- NSF Robert Noyce Scholar ($30,000 full funding); Advanced Certificate in Leadership in Disability and Inclusive Practices.

Rochester Institute of Technology — 2015
Master of Science, Computer Science — GPA 3.2 — *Rochester, NY*

- Concentrations: Artificial Intelligence, Big Data Analytics, Computer Graphics.

Farmingdale State College — 2012
Bachelor of Science, Computer Engineering Technology — GPA 3.3 — *Farmingdale, NY*

- Foundations in computer hardware, embedded systems, and engineering design; early technical training for computational problem-solving.

Research and Professional Experience

Rochester Institute of Technology, Software Engineering Department — Fall 2025 – Present
Graduate Assistant, Quantum and AI Research — *Rochester, NY*

- Build reproducible machine learning evaluation pipelines across local, Colab, and GCP environments with structured logging and validation.
- Develop experimental bandit and decision-making systems for uncertain, heterogeneous settings, with emphasis on reliability and traceability.
- Produce manuscript-ready technical analyses and research artifacts for faculty-led projects.

University of Rochester, Warner School of Education Sep 2024 – Present
CS Teacher Candidate & Education Research (NSF Noyce Scholar) *Rochester, NY*

- Deliver K–12 computer science instruction in active public school placement as part of the NSF Robert Noyce Scholar program, including computational thinking, inclusive pedagogy, and NYS learning standards.
- Conduct applied education research on AI-informed and data-driven approaches to improve learning outcomes for neurodivergent students, applying Universal Design for Learning (UDL) and evidence-based adaptive strategies.
- Produce lesson artifacts and instructional analyses documenting the intersection of AI, inclusive education, and evidence-based CS pedagogy.

Rochester Institute of Technology 2024 – 2025
Equitable Bioinformatics Research — ISTE-780 *Rochester, NY*

- Pioneered fairness assessment applied to RNA secondary structure prediction algorithms, implementing bias-detection frameworks using SHAP analysis and disparate-impact metrics.
- Developed 900+ line modular Python codebase for reproducible fairness-aware evaluation of bioinformatics ML methods.
- Demonstrated statistical significance ($p < 0.01$) in algorithmic bias detection across demographic proxies in biological data.

Rochester Institute of Technology 2025 – 2026
RNA Structure Prediction and Computational Biology (BIO614 and BIO550) *Rochester, NY*

- Led substantial computational work in BIO614: advanced RNA secondary structure prediction by integrating the Nussinov dynamic-programming algorithm with thermodynamic MFE enhancements using Turner parameters and ViennaRNA, achieving >90% accuracy on synthetic motifs.
- Extended prediction pipeline with LSTM and neural-network-based scoring components to capture long-range dependencies in RNA sequences beyond classical dynamic-programming scope.
- Produced formal manuscript-style project (*BIO614-FinalProjectProposal*, Overleaf) with reproducible implementation, structured evaluation, and biological interpretation.
- In BIO550: conducted RNA-seq differential-expression analysis, including reproducible preprocessing, quality-control validation (DESeq2 workflow), and biological interpretation of gene-expression patterns.

VEDADATA Inc. Sep 2022 – Aug 2024
Data Solutions Engineer *New York, NY*

- Designed scalable Python and pandas workflows for ingestion, validation, preprocessing, and analytics operations.
- Strengthened production data reliability through structured quality checks and statistical diagnostics.

VIOME Inc. Jan 2021 – Sep 2022
AI Data Solutions Engineer *New York, NY*

- Developed healthcare-oriented machine learning workflows for preprocessing, feature engineering, model experimentation, and evaluation.
- Supported reproducible analytics workflows in AWS-based environments for predictive health applications.

Selected Technical Writing

- *Big Data Medical Diagnosis: A Multi-Method Approach* (RIT M.S. Computer Science graduate research, 2015) — ensemble and deep learning methods for clinical classification on high-dimensional biomedical data.
- *Predicting Hospital Readmission Rates: A Multi-Method ML Analysis* (DSCI-633 Project Report, 2025) — ensemble and regularized regression methods for clinical risk stratification under distribution shift.
- *BIO614-FinalProjectProposal* (Overleaf manuscript, 2026) — RNA secondary structure prediction with thermodynamic deep learning enhancements, reproducible benchmarking, and biological interpretation.
- *Equitable Bioinformatics: Enhancing Diagnostic Decision-Making through RNA and Biomarker Data* (ISTE-780 Project Report, 2025) — fairness-aware ML for genomic diagnostic algorithms with SHAP-based bias detection and statistical significance testing.
- *Differential Gene Expression in Murine DRG Neurons Following Sciatic Nerve Injury: An NGS Reanalysis* (BIOL550 Computational Biology Project, 2026) — reproducible bulk RNA-seq pipeline using DESeq2, QC validation, and pathway-level biological interpretation of nociceptor gene expression.
- *Fairness-Aware Bandits for Clinical Decision Systems* (DSCI-601 Project Proposal, 2025) — bandit learning with equity constraints applied to high-stakes clinical diagnostic decision-making, addressing demographic disparity in sequential clinical recommendation workflows.

Awards & Recognition

- NSF Robert Noyce Teacher Scholarship (2024–2026) — full funding for M.S. in Teaching Computer Science K–12.
- MS International Scholarship (2012–2015) — MESCYT & RIT, awarded for academic excellence.
- Honorable Mention Technology Award (2012) — NYS STEP/CSTEP, 2nd place capstone project.
- IEEE Alumni Member (2009–Present) — Member No. 90613000.

Technical Competencies

Programming: Python, R, Java, C++, C#, SQL, MATLAB, JavaScript, Bash
Machine Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, model evaluation, uncertainty-aware analysis, fairness-aware ML
Data and Statistics: pandas, NumPy, SciPy, statistical modeling, reproducible benchmarking
Research Tooling: Git/GitHub, Linux, Docker, LaTeX, AWS, GCP

Selected Fit for Uppsala Precision Medicine Position

- Combined computational biology background (BIO614 RNA structure prediction; BIO550 RNA-seq differential expression; ISTE-780 fairness-aware genomic diagnostics) directly matches the group's focus on machine learning for precision medicine and diagnostics.
- Strong Python-based deep learning and reproducibility foundation (PyTorch, TensorFlow, scikit-learn) aligned with the group's image analysis and cancer detection workflows.
- Research motivation to apply fairness-aware and uncertainty-aware ML methods to clinical datasets with direct translational relevance to cancer treatment prediction and biomarker discovery.
- Multidisciplinary profile spanning data science, computational biology, and health AI—suitable for collaboration across MIDA, Centre for Image Analysis, IGP, and SciLifeLab.